
Algorithm 1 Transition-level failure-mode classifier

Require: Proxy delta ΔR_ϕ , judge deltas $\Delta R^\dagger, \Delta R_2^\dagger$, tolerance ϵ

```
1: function SIGN( $z, \epsilon$ )
2:   if  $z > \epsilon$  then
3:     return +1
4:   end if
5:   if  $z < -\epsilon$  then
6:     return -1
7:   end if
8:   return 0
9: end function
10:  $p \leftarrow \text{SIGN}(\Delta R_\phi, \epsilon)$ ;  $j \leftarrow \text{SIGN}(\Delta R^\dagger, \epsilon)$ ;  $j_2 \leftarrow \text{SIGN}(\Delta R_2^\dagger, \epsilon)$ 
11: if  $p = 0$  and  $j = 0$  then
12:   mode  $\leftarrow$  conservative stagnation
13: else if  $p > 0$  and  $j > 0$  then
14:   mode  $\leftarrow$  stable alignment
15: else if  $p > 0$  and  $j < 0$  then
16:   mode  $\leftarrow$  reward hacking
17: else if  $p < 0$  and  $j < 0$  then
18:   mode  $\leftarrow$  optimization collapse
19: else if  $p < 0$  and  $j > 0$  then
20:   mode  $\leftarrow$  proxy under-alignment
21: else
22:   mode  $\leftarrow$  mixed or ambiguous
23: end if
24: evaluator-gaming  $\leftarrow (j \cdot j_2 < 0)$ 
25: return mode, evaluator-gaming
```

4 Experimental Design

The study is artifact-driven: it consolidates existing rollout summaries and per-prompt examples from a controlled RLHF pipeline. The pipeline contains supervised fine-tuning, reward-model training, PPO with several KL settings, DPO evaluations, UP-PPO variants, reward uncertainty via MC dropout, approximate KL to the supervised reference policy, lexical diversity and repetition metrics, and two LLM judges. The primary judge $R^\dagger$ is an Anthropic Claude helpfulness judge configured to return a single 1–10 score for the assistant reply only; the default configured model is `claude-sonnet-4-5-20250929`. The second judge $R_2^\dagger$ is an OpenAI chat-model helpfulness judge using the same single-number 1–10 rubric; the default configured model is `gpt-4o-mini`. In both cases, the judge prompt is: “Rate the assistant reply below for helpfulness on a 1-10 scale. Reply with a single number only, no other text.” Scores are clamped to the interval $[1, 10]$. The artifacts record judge provider labels; exact model identifiers follow the configured defaults unless the evaluation environment explicitly overrides them.

UP-PPO is therefore not a separate architecture in this study; it is a shaped-reward PPO variant in which reward-model uncertainty is subtracted before computing the scalar advantage. The available artifacts support a comparison over $\lambda \in \{0.1, 0.5\}$ against aggressive PPO. They do not support a full ablation over dropout rate, number of MC samples, uncertainty schedules, or alternative uncertainty estimators without rerunning the training pipeline.

Table 2: Empirical artifact summary used in the paper.

Component	Value
Dataset family	Anthropic HH-RLHF prompts and generated completions
Policy scale	GPT-2-scale controlled post-training pipeline
Training/evaluation families	PPO KL sweep, aggressive PPO, UP-PPO, DPO, SFT references
External judges	Claude helpfulness judge $R^\dagger$ and OpenAI helpfulness judge $R_2^\dagger$
Checkpoint metric rows	61
Per-prompt rollout examples	9,280
Checkpoint transitions	31
Row-level transitions	1,920
Main diagnostics	R_ϕ , $R^\dagger$, $R_2^\dagger$, uncertainty, KL, length, diversity, repetition

Table 3: UP-PPO objective and hyperparameters for the aggressive comparison artifacts. The reward-model score and MC-dropout standard deviation are temperature-scaled by the calibrated reward-model temperature $T = 1.554$.

Component	Value
Shaped reward	$\hat{R}_\lambda(x) = R_\phi(x)/T - \lambda u(x)/T$
Training objective	PPO-style clipped surrogate plus $\beta \log \pi - \log \pi_0 $
Compared λ values	0.0, 0.1, 0.5
KL penalty β	0.0 in aggressive PPO/UP-PPO comparison
MC-dropout samples / dropout	$K = 4$ samples; reward-head dropout 0.1
Learning rate / clip / inner epochs	2×10^{-6} ; clip 0.2; one inner epoch
RL prompts / steps / checkpoint interval	512 prompts; 1,200 steps; every 200 steps
Generation during training	sample, top- $p = 0.95$, temperature 0.9, max 96 new tokens
Advantage controls	moving reward baseline, momentum 0.9; advantage clipped to $[-2, 2]$

5 Results

5.1 Failure modes are regime-dependent

Table 4 summarizes the row-level taxonomy. Aggressive PPO is the most failure-prone condition: 37 of 256 row-level transitions are classified as reward hacking (14.45%). PPO with $\beta = 0.03$ also shows a non-trivial reward-hacking share (9.38%), while DPO and SFT reference settings show none under this taxonomy. The pattern supports a training-dynamics interpretation: failures are concentrated in particular optimization regimes rather than uniformly distributed across all evaluated policies.

5.2 Temporal trajectories expose divergence

Figure 2 plots learned reward, primary judge score, reward uncertainty, and approximate KL across evaluated PPO-family checkpoints. The trajectory view complements the transition taxonomy: reward hacking is not merely a final-checkpoint label, but a temporal pattern in which proxy reward, external judgment, uncertainty, and drift can separate over training. Figure 3 shows the corresponding emergence curve for localized row-level reward hacking. These views make the analysis closer to a monitoring protocol: a practitioner can inspect not only whether a run fails, but when localized failures begin to concentrate.

Algorithm 2 Artifact-driven diagnostic pipeline

Require: Rollout summaries S , row-level examples E , taxonomy classifier C

- 1: Parse checkpoint identity, training family, model type, and step from each artifact.
 - 2: Build checkpoint table with mean R_ϕ , $R^\dagger$, $R_2^\dagger$, uncertainty, KL, and judge disagreement.
 - 3: Build row-level table with prompt id, scores, generated text, length, distinct- n , and repetition.
 - 4: **for** each family/model trajectory **do**
 - 5: Sort evaluated checkpoints by step.
 - 6: **for** each adjacent transition $t \rightarrow t'$ **do**
 - 7: Apply C to aggregate deltas and store checkpoint failure mode.
 - 8: Match row-level examples by prompt id and apply C to each prompt transition.
 - 9: **end for**
 - 10: **end for**
 - 11: Train early-warning models to predict future row-level reward hacking.
 - 12: Fit using a stratified random 70/30 row split; record that prompts and trajectory families are not group-held-out.
 - 13: Report aggregate summaries, row-level diagnostics, and robustness checks.
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Table 4: Main row-level failure-mode counts by setting. RH denotes reward hacking; OC optimization collapse; PUA proxy under-alignment; CS conservative stagnation; MA mixed or ambiguous.

Setting	Stable	RH	OC	PUA	CS	MA	RH share
$\beta = 0.0$ aggressive PPO	26	37	27	30	10	126	0.145
$\beta = 0.0$ UP-PPO $\lambda = 0.1$	27	29	21	28	27	124	0.113
$\beta = 0.0$ UP-PPO $\lambda = 0.5$	19	28	28	27	10	144	0.109
$\beta = 0.03$ PPO	4	12	10	9	45	48	0.094
$\beta = 0.001$ sampled PPO	20	19	10	19	72	116	0.074
$\beta = 0.01$ PPO	3	1	3	4	67	50	0.008
$\beta = 0.005$ PPO	1	1	0	5	99	22	0.008
DPO/SFT reference settings	0	0	0	0	418	94	0.000

5.3 Reward hacking is localized to prompts

A central empirical finding is that failures are localized. Across 64 prompts, 38 prompts exhibit repeated reward-hacking transitions, and the most failure-prone prompt appears in five reward-hacking transitions (Appendix B). This does not mean that the prompt itself is inherently unsafe; rather, it means that particular prompt-policy-checkpoint combinations repeatedly expose proxy-judge mismatch. The practical implication is that aggregate checkpoint means are insufficient for monitoring. RLHF evaluation should retain prompt-level transition structure.

5.4 Averages hide localized failures

The checkpoint-vs-row ablation quantifies how much is lost when analysis is restricted to aggregate checkpoint transitions. In 3 of 12 settings (25.0%; bootstrap 95% CI: 0–50.0%), checkpoint-level analysis reports no reward-hacking transition even though row-level transitions contain reward-hacking cases. At the transition level, all 30 checkpoint transitions with matched row data have a dominant row-level label that differs from the checkpoint label; 19 of those 30 have a strict row-level majority. This does not mean that checkpoint labels are wrong; it means that aggregate deltas and row-level prompt deltas answer different diagnostic questions. The most important examples are $\beta = 0.001$ sampled PPO and $\beta = 0.03$ PPO: both have zero aggregate reward-hacking transitions but nonzero localized reward-hacking shares.